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1. Key Component Definitions

Project	Front Brakes
Key Components	Brake Calipers Brake Discs Brake Pads Master Cylinders Reservoir Brake Lines Brake Pressure Sensor IBSe (Brake Booster)

Project	Rear Brakes
Key Components	Brake Calipers Brake Discs Brake Pads Master Cylinder Reservoir Brake Lines Brake Pressure Sensor Residual Pressure Valve Electronic Park Brake Balance Bar

2. Design Factors

2.1 Lifecycle

Calipers
Requires minimal maintenance and likely no replacement. Piston seals (disposable) should be inspected regularly to ensure they are not torn.

Disc
Regular inspection required. Replacement should occur around ~100,000km of regular use. Should be replaced after ~50 track hours* due to heat cycles causing distortion/warping.

Pads
Wear level sensor will be the main indicator of pad wear along with general inspection. Should be replaced after ~50,000km of regular use or ~15 hours of track use*.

Lines
Regular inspection for wear and use of brake pressure sensor.

Master Cylinder
Susceptible to corrosion.

* Metrics based on GT3RS consumable replacement

2.2 Risk

No major changes from Conceptual Design.

Risk	Cause	Response	Strategy
Brake Fade	Brake fluid boiling after absorbing water Brake pads exceeding thermal capacity	Mitigate	Allow for system to cool down in between track usage Inspect and replace brake fluid regularly
Brake Pad/Disc Wear	Expected outcome of consistent brake usage. Accelerated under periods of heavy braking.	Accept	Regular inspections and replacement when required. Brake pad wear detected through wear sensor.
Brake Disc Overheating (Rotor Warping)	Heavy/prolonged braking without allowing for cooling.	Mitigate	Allow for cooling during intensive runs. Choose slotted/ventilated discs for max cooling.
Brake Squeal	Lack of lubrication between pad and disc, or debris.	Avoid	When brakes are serviced/worked on, make sure they are lubricated with grease and checked for debris.
Brake Leak	Worn brake lines, seals, and hoses.	Mitigate	Pressure sensors for each master cylinder should indicate brake leak. Inspect hoses, seals and lines regularly for wear and replace when necessary.
Air in Brake Lines	Using worn down pads, improper caliper maintenance, insufficient fluid in the reservoir.	Avoid	Make sure fluids are topped up at all times, follow maintenance schedules for other parts.

2.3 Innovation

Although it is possible to innovate in terms of brakes, the risk involved with innovation is much higher than the result of innovation – in terms of safety, cost and time. The team would require partnerships with companies already innovating in this area and significant considerations would have to be undertaken in terms of safety to ensure that brakes are compliant with legal standards and user safety.

2.4 Manufacture and Procurement

Due to the high risk and crucial nature of the braking system, strict procedures would have to be followed if brakes were to be manufactured. Therefore, all brake components will be bought off the shelf.

Component	Product Name	Purchase Source	QTY/CPU	Cost (Total)
Brake Calipers (Front)	AP Racing CP9200	Demon-Tweeks	2 / $990	$1980
Brake Calipers (Rear)	AP Racing CP9202	999autoshop	2 / $2073.33 (pair)	$2073.33
Brake Discs	AP Racing CP3047/252-253	Demon-Tweeks	4 / $746.44	$2985.76
Brake Pads	AP Racing CP3215D50 (APF403/405)	CompFriction (unlisted)	4 / ?	
Master Cylinder	AP Racing CP7854	Murray Motorsport	2 / $622.79	$1245.58
Reservoirs	AP Racing CP2293-141/-143	RPXpress	2 / $88.37	$176.74
Brake Lines	Stainless Steel Brake Lines 3/16" (7.62m)	Aeroflow Performance	? / $79.29	
Brake Hose	Proflow 350mm AN3 Braided Brake Hose	Proflow	? / $88.25	
Brake Line Pressure Sensor	Pressure Sensor 0-1500PSI	Microtechefi	2 / $125	$150
Residual Pressure Valve	Residual Pressure Valve 2PSI	Proflow	2 / $21.22	$42.44
Electronic Park Brake (Left)	Wilwood 120-15703-BK	CarModsAustralia	1 / $764.95	$764.95
Electronic Park Brake (Right)	Wilwood 120-15702-BK	CarModsAustralia	1 / $709.95	$709.95
Balance Bar	AP Racing CP5520-25LC	CompFriction (unlisted)	1 / ?	?
Brake Booster	IBSe9	Already owned!	2 / $20,000	$40,000
			TOTAL	$52,311.79

Commented [MK1]: Buy integrated disc with mounting bell?

Commented [MK1R2]: Manufacturing vs safety

Commented [MK2]: Materials:
APF403 Front
APF405 Rear

2.5 Design and Production Plan

2.6 Cost Analysis

Costs for each individual component is provided in 2.4 – Manufacture and Procurement.

Actual Cost –

Cost to Sunswift –

2.7 Educational outcome

Although there is no component manufacturing for brakes, there is still significant educational value in terms of materials and the braking system itself. This project will educate students about the part of Vehicle Dynamics, and Mechanical as a whole, about the assembly of a braking system and its various components, as well as the choice of materials for each component and how it impacts performance and reliability. The procurement also allowed for research of various styles of parts, such as master cylinders, and where to source them from. Once assembled, students will be able to analyse the impact of different setups on braking performance and work with other departments to improve efficiency and effectiveness.

2.8 Upskilling Development

Although no manufacturing upskilling may be required, there will be an increase in knowledge in both braking data and the assembly of braking systems and how it integrates with the rest of a vehicle.

3. Requirements

3.1 ADR Requirements

ADR 31 – Brake System for Passenger Cars

ADR 42 – General Safety Requirements

10.	**BRAKE HOSE AND BRAKE TUBING**
10.1.	Flexible hydraulic brake hoses, air and vacuum brake tubing, air and vacuum hose, and flexible and hydraulic power hose, between the *'Brake Power Unit 31/00'* or *'Brake Power Unit 35/...'* and the master cylinder or its equivalent must: (a) conform to any edition of the United States Federal Motor Vehicle Safety Standard (FMVSS) No. 106 (49.CFR.571.106 – Brake Hoses) in force between 1 October 2010 and the commencement of this rule, or to an applicable SA, SAE, BS, JIS, DIN, ISO or UN standard for flexible brake hoses, air brake tubing or hose, vacuum brake tubing or hose, or hydraulic power tubing or hose, current between 1 January 2005 and the commencement of this rule;[1] and (b) be fitted/routed so as to prevent chafing, kinking or other mechanical damage under normal motion of the moving vehicle parts.

ADR 88 – Stability Control (SEV Exempt)

ADR 89 – Brake Assist (SEV Exempt)

ADR 98 – Advanced Emergency Braking for Passenger Vehicles and Light Goods Vehicles (SEV Exempt)

3.2 Functional Requirements

The brakes are, under normal circumstances, designed to slow down and stop the vehicle. This works through clamping down on the brake disc.

3.2.1 Abnormal Operation: Emergency Scenarios

In emergencies, brakes can be operated differently to suit the scenario. For example, the ABS will help prevent a lock up of the wheels in cases where the braking force exceeds static friction. Pedestrian sensors, crash avoidance?

3.2.2 Abnormal Operation: User Error

N/A

3.2.3 Testing/Maintenance/Fault Diagnostic

The brakes will have to be tested prior to heavy usage. This can be done through measuring the stopping distance or a decelerometer to measure G-forces. Brakes will have to be maintained regularly, through checking brake pad wear (sensor), brake disc wear, brake line condition, and the condition of the hydraulic braking fluid. Faults can be diagnosed through the OBD2 port (?) and through looking at brake traces and temperatures.

3.3 Performance Requirements

3.3.1 Teamwide Specifications

Regen braking?

3.3.2 Technical Specifications

Maximum longitudinal G-force during braking is approximately 1.3G.

3.3.3 Safety Specifications

Temperature

3.4 Interface Requirements

Interior – Pedal Box, Pedal Placement

3.5 Reliability Requirements

It is crucial that the brakes are reliable. **Brake fade** must be mitigated through various methods (friction fade, mechanical fade, fluid fade). Significantly, fluid fade occurs when the brake fluid starts boiling due to hard brake usage. Brake fluid must be changed regularly due to its hygroscopic properties (reduces boiling point). Pad fade

can occur when the pads exceed thermal capacity and stopping distance is greatly increased.

Redundancy is integrated through two master cylinders in case of failure of either braking axle.

3.6 External Requirements

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